

Software Test Plan (STP)

Smart Complaint Management System (SCMS)

Project: Smart Complaint Management System (SCMS)

Version: 1.0

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1. Introduction

Purpose: This document is the Software Test Plan (STP) for the Smart Complaint Management System (SCMS) v1.0. It explains what we plan to test, how we plan to test it, who is responsible for each part, and the schedule we are following, so that SCMS meets the functional and non-functional requirements written in our SRS.

Scope: Testing covers the citizen-facing complaint lodging and tracking pages, the admin dashboard used for triage, assignment and resolution, SLA-based auto-escalation, and email/SMS/in-app notifications. Payment processing, third-party ticketing system integration, and a native mobile app are excluded, since they are not part of our project.

References: SCMS Software Requirements Specification (SRS) v1.0, OWASP Top 10.

Definitions:

- SCMS – Smart Complaint Management System
- OTP – One-Time Password
- RBAC – Role-Based Access Control
- SLA – Service Level Agreement
- JWT – JSON Web Token
- MVC – Model-View-Controller
- CSRF – Cross-Site Request Forgery
- XSS – Cross-Site Scripting
- RTM – Requirements Traceability Matrix

2. Test Items

- AuthModule (Registration & Authentication)
- ComplaintModule (Complaint Lodging)
- TrackingModule (Complaint Tracking & Status Updates)
- AdminModule (Admin Dashboard, Assignment & Resolution)
- NotificationModule (Email/SMS/In-app Notifications)
- ReportingModule (Analytics & Reporting)
- DataLayer (Data Persistence & Encryption)

3. Features to be Tested

Features mapped to SRS requirement IDs:

- SCMS-F-001 to SCMS-F-004: Citizen/Admin registration and login (OTP verification, checking credentials, password hashing, account lockout after failed attempts)
- SCMS-F-010 to SCMS-F-013: Complaint lodging (category/description/location, unique Complaint ID generation, checking mandatory fields, uploading photo evidence)
- SCMS-F-020 to SCMS-F-024: Complaint tracking and status updates (tracking by ID or account, status timeline, notifications, reopening a complaint)
- SCMS-F-030 to SCMS-F-035: Admin resolution and assignment (dashboard view, filter/sort, assigning complaints, updating status with remarks, SLA auto-escalation, reassignment)
- SCMS-F-040 to SCMS-F-042: Notifications, feedback and reporting (status-change alerts, citizen rating/feedback, category/department-wise reports)
- SCMS-NF-001 to SCMS-NF-006: Response time, uptime, encryption of PII, responsive UI, scalability, and a modular codebase
- SCMS-SR-001 to SCMS-SR-005: TLS enforcement, role-based access control, input sanitisation, tamper-evident audit logging, and CSRF protection

4. Features Not to be Tested

- Native mobile application interfaces – SCMS is a web-only project, no native app is in scope
- The internal workings of the third-party SMS/Email gateway – we treat it as an external dependency and mock/stub it during testing
- Hosting infrastructure limits beyond what our application-level scalability tests already cover

5. Test Approach / Strategy

Levels:

- Unit tests (module-level: AuthModule, ComplaintModule, TrackingModule, AdminModule, NotificationModule, ReportingModule)
- Integration tests (module-to-module and module-to-database interactions)
- System tests (end-to-end SCMS flows for both citizen and admin)
- Acceptance tests (UAT with a few representative citizen and admin users)

Types:

- Functional testing (core features listed in Section 3)
- Regression testing (after every fix and before every release)
- Performance testing (response time and load, up to 500 concurrent users)
- Usability testing (UI clarity, responsiveness and accessibility from 360px to 1920px)

Entry Criteria: A stable build is delivered, test data is ready, the test environment is set up, and the RTM is aligned with the SRS.

Exit Criteria: All planned test cases are executed, there are 0 open critical defects, every High-priority functional and security requirement is verified, and all acceptance criteria are met.

5.1 Security Validation

SCMS stores personal citizen data (name, contact details, address, complaint photos), so security testing covers:

- **Security Requirements Traceability:** Each security requirement (SCMS-SR-001 to SCMS-SR-005) is linked to a test case in the RTM (SRS Section 8).
- **Validation Objectives:** Confirm TLS is enforced everywhere, passwords are hashed and never stored/logged in plain text, a Complainant account cannot reach Admin routes, inputs are sanitised against SQL Injection and XSS, audit logs cannot be silently edited, and CSRF tokens are required on state-changing requests.
- **Validation Methods:** OWASP ZAP scans for SQLi/XSS, manual TLS handshake/certificate checks, password hash inspection (bcrypt/argon2), JWT/session tampering checks, manual RBAC route testing, and CSRF token checks.
- **Ownership:** The Test Engineer (Harsh Jha) runs the security test cases, the QA Lead (Chetan Jevoor) reviews results before sign-off, and the Dev Lead (Harsh Gupta) fixes issues found.
- **Timing:** Security checks are run once a module is stable in system testing, and again just before UAT sign-off.
- **Tools & Environment:** OWASP ZAP and Postman, run against the staging test environment described in Section 6.

6. Test Environment

Hardware: Standard desktop, laptop and smartphone devices with a camera (for photo evidence capture) and an internet connection. No special hardware is required.

Software: SCMS application (Python Flask/Django, or PHP Laravel backend), MySQL/PostgreSQL test database, modern web browsers (Chrome, Firefox, Edge, Safari).

Tools: Postman (API testing), Selenium (UI automation), JMeter (performance/load testing, up to 500 concurrent users), Jira (defect and requirement tracking), OWASP ZAP (security/penetration testing).

Test Data: Dummy citizen and admin accounts, sample complaints across different categories, mock photo evidence files, and a few simulated SLA-breach scenarios.

7. Test Schedule

Milestones:

- Test case design: 05-Sep-2025
- Environment setup: 07-Sep-2025
- Test execution start: 08-Sep-2025
- Test execution end: 20-Sep-2025
- UAT: 22-Sep-2025 to 25-Sep-2025

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated)
- Test Scripts
- Test Data
- Test Execution Logs

- Defect Reports (Jira)
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Chetan Jevoor	Prepares the plan, coordinates test execution, signs off on test readiness
Test Engineer	Harsh Jha	Designs and executes test cases, logs defects, keeps the RTM updated
Developer / Dev Lead	Harsh Gupta	Fixes defects, helps with triage, reviews code changes
Product Owner	Chinnala Manish Ranga	Approves test results, signs off on release readiness

10. Risks and Mitigation

Risk	Mitigation
High concurrent traffic causing database locks	Run JMeter load tests up to and beyond 500 concurrent users before release
Notification gateway latency (email/SMS)	Mock the external email/SMS APIs during integration tests
Delay in stable build delivery	Ask the dev team for early smoke builds
Test environment downtime	Keep a backup environment ready on a cloud VM

11. Assumptions & Dependencies

- The Python (Flask/Django) or PHP (Laravel) backend and the relational database (PostgreSQL/MySQL) will be stable and available for testing.
- Test data (citizen/admin accounts, sample complaints) will be ready before execution begins.
- Email/SMS gateway credentials or mocks will be available for notification and OTP testing.
- Source code and requirements will stay tracked on GitHub and Jira throughout testing.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours
- The build is too unstable and blocks more than 30% of the planned test cases
- A critical security vulnerability is found

Resume testing if:

- The blocking defects are fixed and verified in a new build
- The test environment is stable again and smoke tests pass

13. Test Case Management & Traceability

The RTM (SRS Section 8) ensures mapping of SCMS requirements to test cases.

Example:

- SCMS-F-001 (Citizen registration with OTP) → TC-Reg-01
- SCMS-F-002 (Authenticate registered user) → TC-Auth-01
- SCMS-F-003 (Hash and salt stored passwords) → TC-Sec-01
- SCMS-F-004 (Lock account after failed logins) → TC-Auth-02
- SCMS-F-010 (Lodge new complaint) → TC-Lodge-01
- SCMS-F-011 (Auto-generate Complaint ID) → TC-Lodge-02
- SCMS-F-020, SCMS-F-021 (Track complaint by ID / account) → TC-Track-01, TC-Track-02
- SCMS-F-030 (Assign complaint to officer) → TC-Admin-01
- SCMS-F-034 (Auto-escalate overdue complaints, SLA) → TC-Admin-05
- SCMS-NF-001 (Dashboard response time target) → TC-Perf-01
- SCMS-NF-003 (Encryption of PII) → TC-Sec-02
- SCMS-SR-001 (TLS 1.2+ mandatory on all connections) → TC-Sec-03
- SCMS-SR-002 (Role-based access control) → TC-Sec-04
- SCMS-SR-003 (Input sanitisation, SQLi / XSS) → TC-Sec-05
- SCMS-SR-004 (Tamper-evident audit log) → TC-Sec-06
- SCMS-SR-005 (CSRF protection on state-changing requests) → TC-Sec-07

14. Test Metrics & Reporting

Metrics collected:

- % of test cases executed
- % passed / failed
- Defect density (per module: AuthModule, ComplaintModule, TrackingModule, AdminModule, NotificationModule, ReportingModule, DataLayer)
- Defect aging
- Requirement coverage (against SRS RTM, tracked as N/P/A)

Reports:

- Daily execution status
- Final Test Summary Report (submitted to the Product Owner for sign-off)

15. Approvals

Role	Name	Signature / Date
QA Lead	Chetan Jevoor	
Dev Lead	Harsh Gupta	

Product Owner	Chinnala Manish Ranga	
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